

Hadoop Ecosystem and Yarn

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1. Hadoop Ecosystem Components

The Hadoop Ecosystem is a collection of open-source projects that work together to provide a complete Big Data solution. The major components are:

A. Core Components

- HDFS (Hadoop Distributed File System)
- MapReduce
- YARN (Yet Another Resource Negotiator) management

Component	Function
Hive	Data warehouse infrastructure; converts SQL-like queries into MapReduce jobs.
Pig	A high-level scripting language for creating complex data transformations that can be executed on Hadoop.
HBase	NoSQL database built on top of HDFS; column-oriented and supports random read/write.
Sqoop	Transfers data between Hadoop and relational databases (import/export).
Flume	Collects, aggregates, and moves large amounts of log data into HDFS.
Oozie	Workflow scheduler for Hadoop jobs.
Zookeeper	Distributed coordination service.
Mahout	Machine learning library on Hadoop.
Ambari	Cluster management tool for monitoring, and managing Hadoop clusters.

Table: Supporting Components in Hadoop Ecosystem

2. Hadoop Schedulers

Types of **Hadoop** Schedulers



Figure: Hadoop scheduler

Hadoop 2.0 – New Features

Hadoop 2.x was a major upgrade over Hadoop 1.x, solving scalability, fault-tolerance, and flexibility issues. The key improvements include:

- NameNode High Availability
- HDFS Federation
- MapReduce v2 (MRv2)
- YARN

NameNode High Availability (HA):

- Introduced Active/Standby NameNodes to eliminate SPOF(Single Point of Failure).
- Two NameNodes run in the cluster:
 1. Active NameNode – Handles all client requests.
 2. Standby NameNode – Maintains a synchronized copy of metadata and takes over if Active fails.

HDFS Federation :

Introduced to address the memory limitations of a single NameNode, HDFS Federation allows for the horizontal scaling of the NameNode service.

- Federation allows multiple independent NameNodes to manage different parts of the namespace.
- Each NameNode has its own Namespace Volume.
- DataNodes register with all NameNodes, but store blocks for multiple namespaces.

MapReduce v2 (MRv2) :

Separates resource management from job execution.

JobTracker is split into:

- ResourceManager (RM)– Allocates resources across the cluster.
- ApplicationMaster (AM)– Manages job execution for a single job.

YARN (Yet Another Resource Negotiator)

YARN is the biggest innovation in Hadoop 2.x and serves as the cluster operating system.

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Criteria	Hadoop 1.x	Hadoop 2.x
Architecture	Master-slave	Multi-node cluster setup
Resource management	JobTracker for resource management and scheduling	Yet Another Resource Negotiator (YARN) for resource management
Number of NameNodes	Single NameNode	Multiple NameNodes (high availability)
Scalability	Supports up to 4,000 nodes	Supports more than 10,000 nodes
Fault tolerance	Limited to secondary NameNode	NameNode high availability with automatic failover
Job scheduling and tracking	JobTracker manages both	Resource Manager for scheduling and Application Master for tracking
MapReduce	Tightly integrated	Just one of the many applications under YARN
Data processing model	Only MapReduce	Supports various data processing models like graph processing and others
Cluster utilisation	Underutilised in some cases	Better utilisation due to multiple workloads and generic resource model

Figure: Difference between Hadoop 1.x and Hadoop 2.x